

Predicting 30-day Readmission in Diabetic Patients

Panagiotis Giadikiaroglou - Connor Wang - Panagiotis Galanis - Dimitris Deliakidis

Why This Matters

CMS

Penalties under the Hospital
Readmissions
Reduction Program (HRRP)

30-day

The clinical & regulatory window
for readmission tracking

Targeted Care

Follow-up calls, medication
reconciliation, discharge education

Problem

Hospitals face direct financial penalties from CMS for excess 30-day readmissions. Identifying high-risk patients before they leave is a clinical and operational priority.

Stakeholders

Hospital administrators · clinical care teams · patients

Value Proposition

A model that flags high-risk patients at discharge lets hospitals concentrate transitional-care resources where they have the highest expected impact.

Problem Statement & Dataset

- Dataset

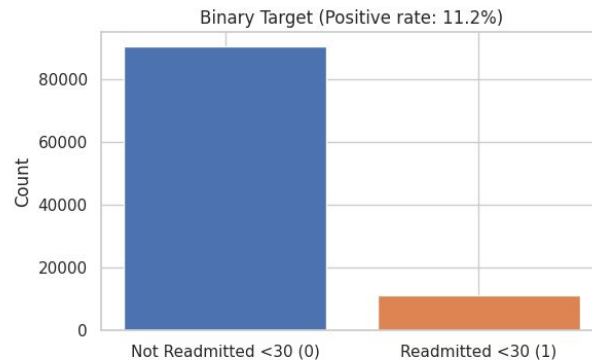
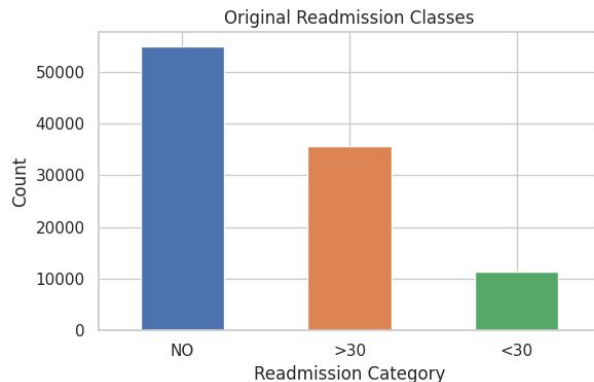
- 130 US hospitals
- **101,766 patient encounters | 50 features**
- Features: demographics, diagnoses, medications, lab results, prior utilization
- 3 ID tables joined via DuckDB

- Goal

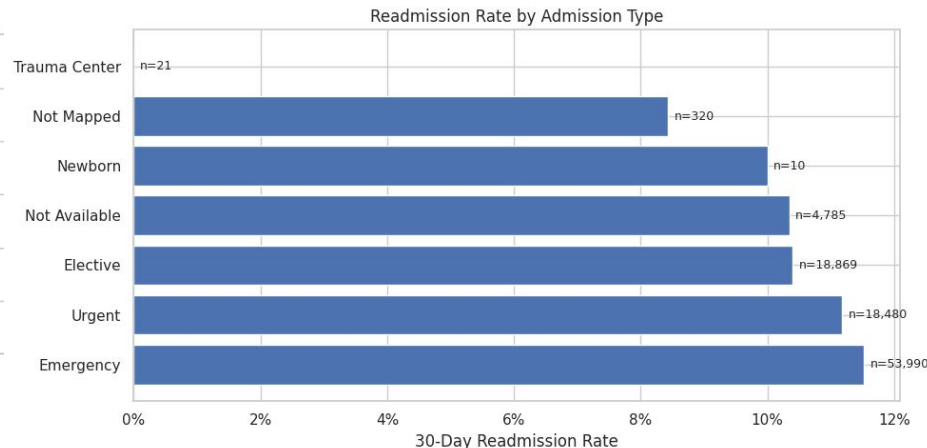
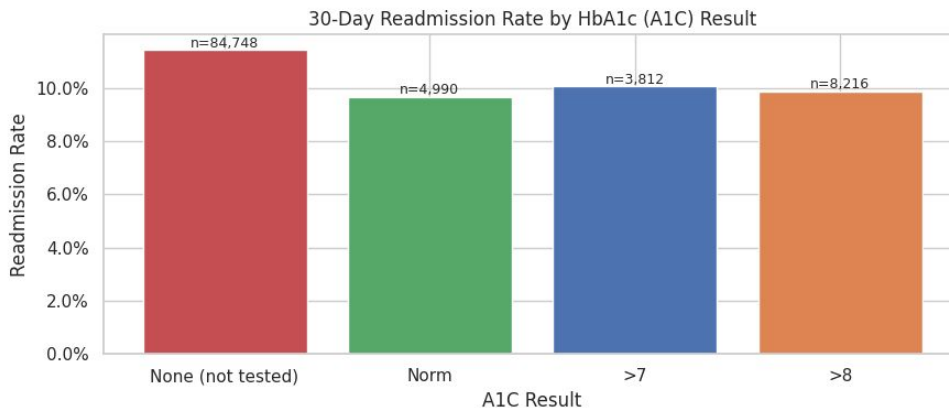
- **Binary Classification:**
readmitted <30 days → 1,
else → 0

! Class Imbalance

- Only 11.2% positive class
- Accuracy is misleading metric.
- Primary metrics:
ROC-AUC and **F1-score**



Exploratory Data Analysis

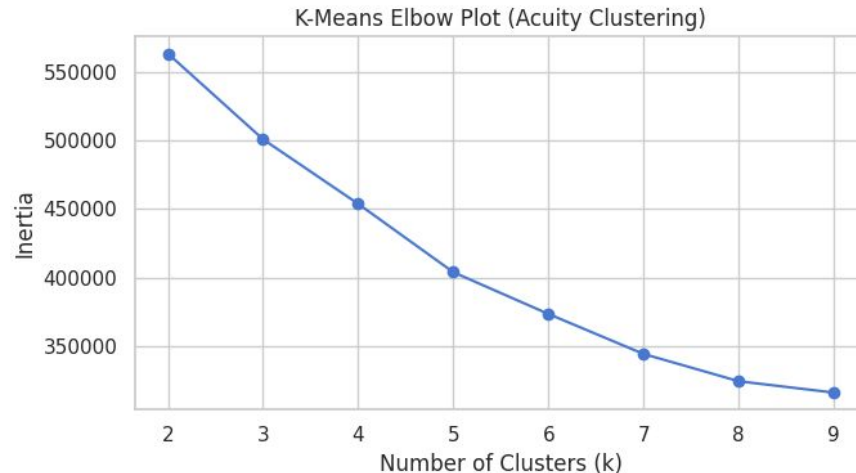


Key Findings

- HbA1c NOT tested → 11.42% readmission vs. 9.85% when tested (→ Hypothesis 1)
- Emergency admissions: highest readmission rate (11.79%), non-emergency: 10.88% (→ Hypothesis 2)
- Strongest numeric predictor: number_inpatient ($r = 0.168$) – prior utilization dominates

Data Preprocessing & Feature Engineering

- Dropped / Flagged
 - Weight (96.9% missing), payer_code (40%), medical_specialty (49%) → **dropped**
 - A1Cresult / max_glu_serum → **binary measured flags**
 - Non-at-risk dispositions **removed** (expired, hospice)
- Feature Engineering
 - ICD-9 codes: ~900 unique → 17 **clinical categories**
 - **total_prior_visits** = inpatient + outpatient + emergency
 - **num_drug_changes**: count of medication adjustments
 - **K-Means** (k=4): acuity_cluster feature (train set)
- Train / Test Split
 - **GroupShuffleSplit** on patient_nbr – 80 / 20
79,456 train | 19,800 test (no patient leakage)
 - **SMOTE** inside ImbPipeline (training folds only)



Cluster	n	Readmit Rate	Avg LOS	Avg Meds	Avg Prior Visits
3	26,495	8.7% (lowest)	3.0 days	11.3	0.64
0	13,372	9.1%	3.3 days	16.7	0.67
1	40,911	12.8%	3.6 days	14.4	1.68
2	18,565	13.7% (highest)	8.8 days	25.8	1.32

ICD-9 Diagnosis Grouping

~900 unique ICD-9 codes → 17 broad clinical categories

Approach

- Regex pattern matching on the official ICD-9-CM chapter ranges (CDC)
- 17 broad clinical categories (Circulatory, Endocrine, Respiratory, ...)
- E- and V-prefix codes grouped into Other

Most common primary diagnoses

- Circulatory — 29,585 (29.7%)
- Endocrine / Metabolic — 11,304 (11.4%)
- Respiratory — 9,919 (9.9%)

Highest 30-day readmission rates

- Blood Disorders — 13.6%
- Endocrine / Metabolic — 12.9%
- Infectious — 12.3%
- Mental Disorders — 12.2%

Lowest risk:

Pregnancy (6.1%) · Musculoskeletal (9.5%)

K-Means Patient Acuity Profiles

k = 4 (elbow method) · trained on training-set only · cluster label appears in XGBoost top-15 feature importance

Cluster	n	Readmit Rate	Avg LOS	Avg Meds	Profile
3	26,495	8.7%	3.0 d	11.3	Low acuity · simple cases
0	13,372	9.1%	3.3 d	16.7	Moderate acuity · low prior use
1	40,911	12.8%	3.6 d	14.4	Moderate · high prior utilization
2	18,565	13.7%	8.8 d	25.8	High acuity · long stays

Takeaway: Cluster 2 (high acuity) and Cluster 1 (high prior utilization) carry ~50% higher readmission rates than Cluster 3 — a clear, unsupervised stratification.

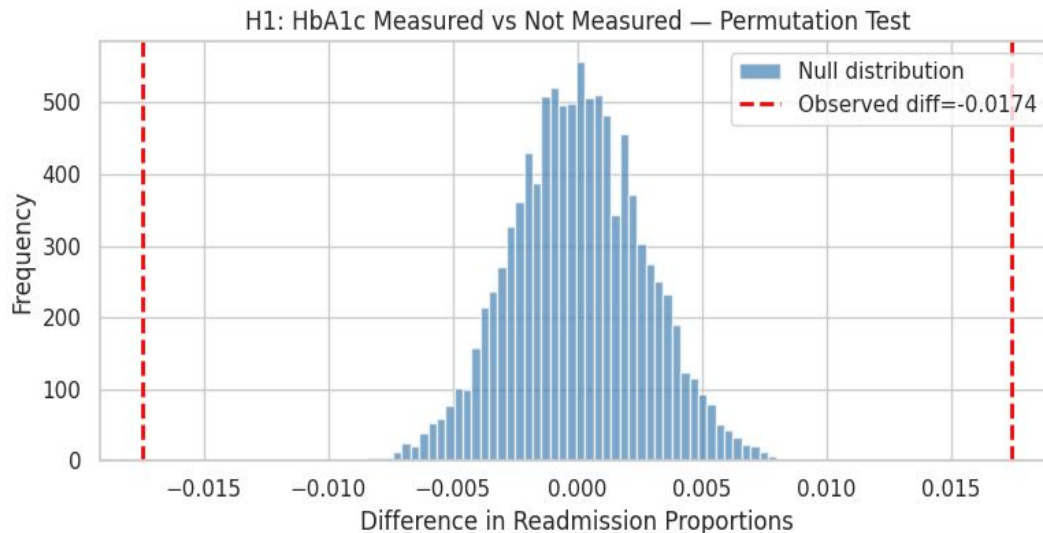
Hypothesis 1: HbA1c Measurement & 30-Day Readmission

- Hypothesis

- H_0 : Readmission rate is the same whether HbA1c was measured or not

- Results

- HbA1c tested: 9.85% readmission
- HbA1c NOT tested: 11.42% readmission
- Observed diff: -1.57 percentage points
- **P-value < 0.0001** → **REJECT H_0 at $\alpha=0.05$**



- Clinical Interpretation

- HbA1c testing is a proxy for proactive diabetes management. Better-managed patients have lower readmission risk (not a direct causal effect). Supports including `a1c_measured` (binary) as a feature.

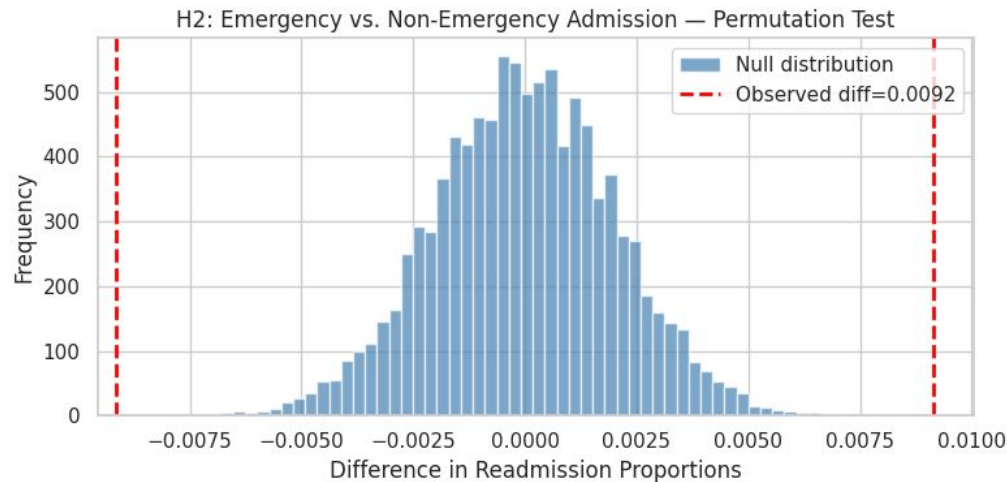
Hypothesis 2: Emergency vs Non-Emergency Readmission

- Hypothesis

- H_0 : The 30-day readmission rate is the same for Emergency and Non-Emergency admissions.

- Results

- Emergency: 11.79% readmission (95% CI: [11.51%, 12.07%])
- Non-Emergency: 10.88% (95% CI: [10.60%, 11.16%])
- Observed diff: 0.92 percentage points
- **P-value < 0.0001 and CIs don't overlap → REJECT H_0 at $\alpha=0.05$**



- Clinical Interpretation

- Emergency admissions, often indicating a more acute and less stable patient presentation, are associated with a higher likelihood of 30-day readmission. Supports including admission type as a feature.

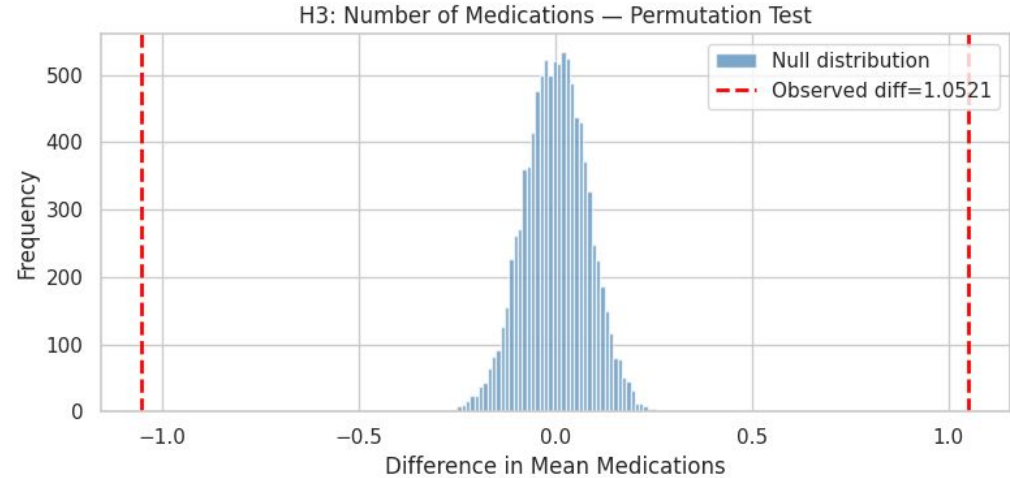
Hypothesis 3: Number of Medications and Readmission

- Hypothesis

- H_0 : The average number of medications is the same for readmitted and non-readmitted patients.

- Results

- Re-admitted mean: 16.91 medications
- Non-readmitted mean: 15.86 medications
- Observed diff: 1.05 medications
- **P-value < 0.0001** → **REJECT H_0 at $\alpha=0.05$**



- Clinical Interpretation

- Patients readmitted within 30 days tend to be on a higher average number of medications, possibly suggesting they are inherently sicker or have more comorbidities, increasing risk of readmission. This finding supports including number of medications as a feature.

Machine Learning Pipeline

- Patient-Level Train/Test Split
 - **Data leakage:** many patients have multiple encounters across 10 year period
 - To solve this we use GroupShuffleSplit on the patient_nbr feature, which is a unique identifier
- K-Means Acuity Clustering (Unsupervised Learning)
 - We use K-means to create patient acuity_cluster labels, which serves as an engineered feature for the model to evaluate
 - Patients are grouped into four acuity profiles ($k = 4$)
- Synthetic Minority Oversampling
 - To address the imbalance (only 11.2% positive class), we use SMOTE to interpolate synthetic data for the minority class to prevent models from always predicting negative class
- Cross Validation
 - Used 3-fold cross validation with imblearn.pipeline, which applies SMOTE only to training folds, ensuring synthetic data does not corrupt validation folds
- Model Selection and Hyperparameter Optimization
 - We use RandomizedSearchCV to fine tune the settings for each model, optimizing for ROC-AUC score
 - Compared Logistic Regression, RandomForest, XGBoost for performance

Randomized Search Results and Evaluation

- Running RandomizedSearchCV for 20 iterations, using 3 fold cross-validation, scored on ROC-AUC leads to the above results
- XGBoost model achieves highest score, followed by Logistic Regression and Random Forest
- Parameters: learning_rate=0.1, max_depth=7, n_estimators=300, subsample=0.8

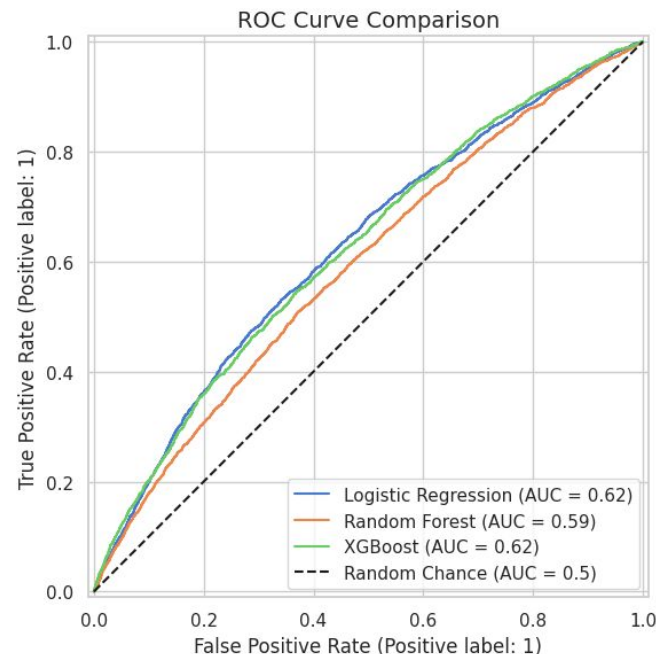
Model	Best CV ROC-AUC	Best Parameters
Logistic Regression	0.6178	C=10
Random Forest	0.5966	n_estimators=200, max_depth=20, max_features=0.3, min_samples_split=10
XGBoost	0.6192	n_estimators=300, learning_rate=0.1, max_depth=7, subsample=0.8

- A moderate learning rate with 300 rounds suggests the boosting ensemble benefits from more numerous and gradual iterations
- max_depth=7 is deeper than typical boosting defaults (3-5), reflecting that this dataset has complex interactions the shallow trees alone cannot resolve.
- Row subsampling (subsample=0.8) adds regularizing stochasticity (i.e. prevent overfitting by only allowing subset of data to train particular tree)

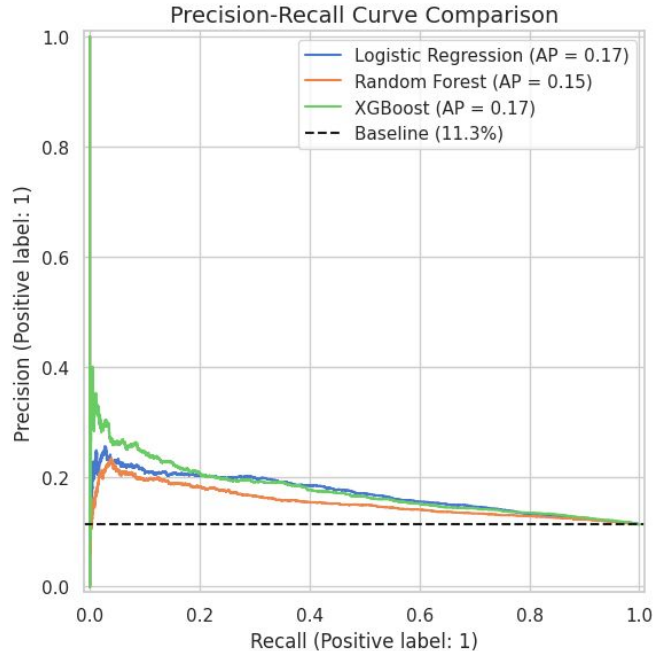
Model Evaluation and Visualization

ROC-AUC (Receiver Operating Characteristic - Area Under Curve): Measures the model's ability to rank a randomly chosen readmitted patient higher than a randomly chosen non-readmitted patient (True Positive vs False Positive)

- XGBoost and Logistic Regression almost tied for highest AUC of 0.62, with XGBoost just ahead.
- The Random Forest model underperformed with an AUC of 0.59, sitting visibly closer to the random chance line. This suggests it struggled to generalize to the unseen test data, likely overfitting to the synthetic (SMOTE) examples generated during training.
- While all three models beat random chance, an AUC of 0.62 indicates a modest predictive signal. Predicting 30-day readmissions is notoriously difficult, as patient behavior post-discharge is influenced by countless unmeasured factors (like socioeconomic status, diet, and home support).
- To catch a high percentage of true readmissions (y-axis), the hospital must accept a high false alarm rate (x-axis), meaning wasting resources on patients who did not need it



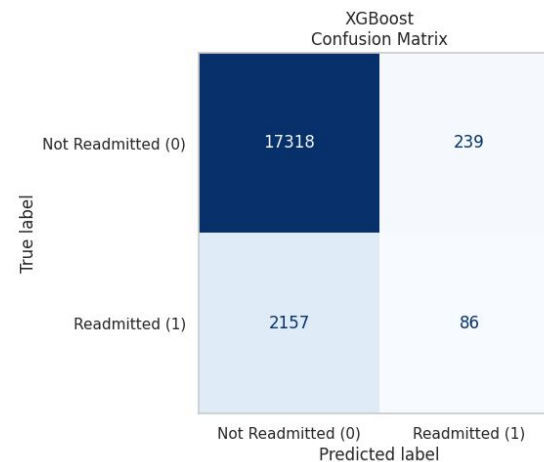
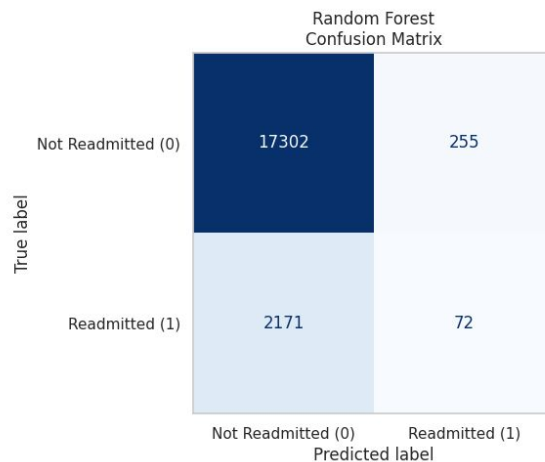
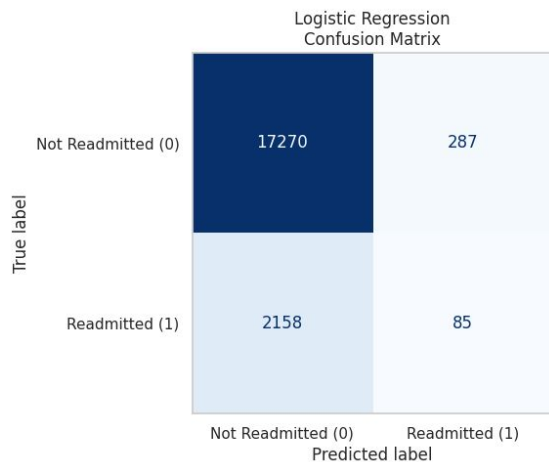
Model Evaluation and Visualization



Precision-Recall (PR) Curve: Often more informative than ROC for highly imbalanced datasets, as it directly evaluates the trade-off between capturing the minority class (Recall) and avoiding false alarms (Precision).

- **XGBoost & Logistic Regression Lead:** Both models tied for the best performance with an Average Precision (AP) of 0.17, successfully outperforming the 11.3% baseline (represents random guessing based on the dataset's class imbalance).
- **Random Forest Struggles:** Random Forest lagged slightly behind (AP = 0.15), reinforcing that it struggled to generalize to unseen patients after training on synthetic (SMOTE) data.
- **The Operational Trade-off:** The steep downward slope of the curves illustrates the core challenge of predicting readmissions. To successfully catch a large percentage of true readmissions (high recall), the hospital must accept a high volume of false alarms (low precision).

Model Evaluation and Visualization



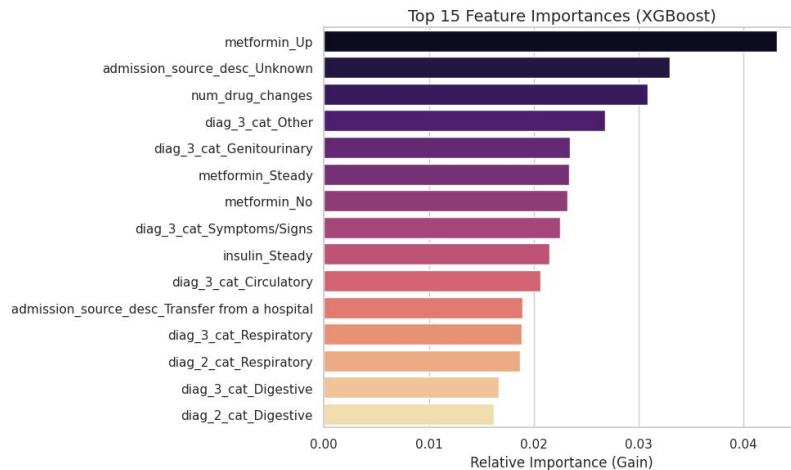
Confusion Matrices: Provides a raw count of True Positives, False Positives, True Negatives, and False Negatives.

Model Evaluation and Visualization Summary

Model	ROC-AUC	F1-Score (positive)
Logistic Regression	0.6247	0.0650
Random Forest	0.5920	0.0560
XGBoost	0.6227	0.0700

F1-Score: The mean of Precision and Recall, giving us a single metric for the minority class performance.

XGBoost Feature Importance — Top 15



Prior utilization dominates

number_inpatient and engineered total_prior_visits lead.

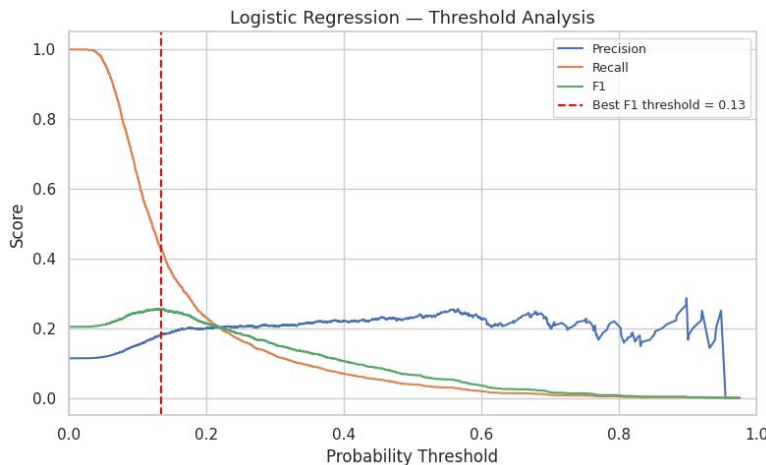
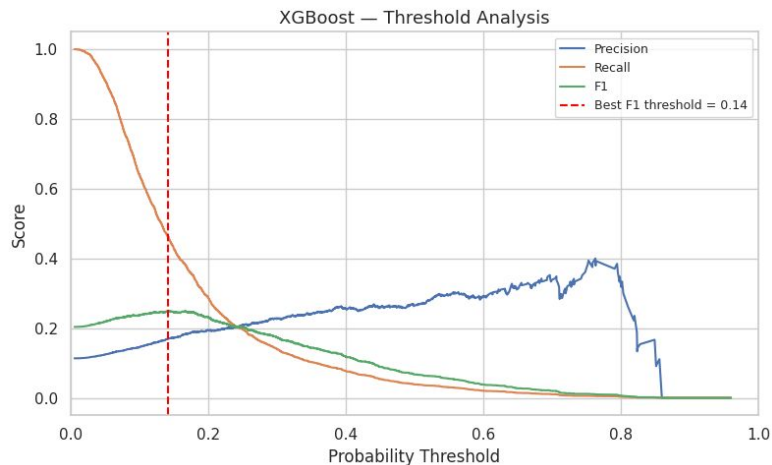
K-Means cluster adds signal

acuity_cluster appears in the top 15.

Systemic patterns > single labs

HbA1c and glucose rank below utilization.

Threshold Analysis



- 50% Default is Ineffective: Using a standard 0.5 probability threshold yields near-zero recall. It fails to flag patients due to the severe 11% class imbalance in the dataset.
- Prioritizing Recall Over Precision: In a clinical setting, missing a high-risk patient is much more costly than making an unnecessary follow-up call. We must lower the alarm threshold to catch more true readmissions.
- The Clinical Sweet Spot (~0.15): Lowering the risk threshold to 15% aligns with the mathematical optimum for F1-scores and provides a highly actionable screening tool.
- The Operational Trade-off: At a 0.15 threshold, our XGBoost model successfully catches ~43% of all true readmissions.
- Managing False Alarms: This threshold yields a precision of 17–19%, meaning roughly 1 out of every 5 or 6 flagged patients will actually be readmitted.
- Recommendation: The final deployed threshold should be calibrated directly to the hospital's specific financial ratio: the cost of a CMS readmission penalty versus the cost of a preventative phone call.

Polars: Discharge Disposition Risk Profile

- We use Polars to perform a fast group-by aggregation summarising readmission risk and patient complexity by discharge disposition. Discharge disposition was one of the strongest signals in the EDA, and here we quantify it across multiple dimensions simultaneously.
- Critical Risks: Found that patients transferred to Skilled Nursing Facilities (SNFs) represent the largest high-risk group (14.7% readmission rate).
- Actionable Strategy: Established that post-discharge destination is a primary predictor of risk, providing a roadmap for targeted hospital intervention programs.

discharge_dispositi on_desc --- str	readmit_rate --- f64	n_encounters --- u32	avg_lo s --- f64	avg_medications --- f64	avg_prior_visits --- f64
Discharged/transfer red/referre...	0.3669	139	3.83	14.51	1.33
Discharged/transfer red to anot...	0.277	1993	5.79	20.32	0.95
Discharged/transfer red to anot...	0.2086	1184	5.65	16.18	0.82
Discharged/transfer red to anot...	0.1607	2128	3.86	15.56	1.32
Discharged/transfer red to SNF	0.1466	13954	5.79	17.96	1.41
...	...	...	...	...	...
Discharged/transfer red to a lo...	0.0728	412	7.06	23.43	1.47
Discharged/transfer red to a fe...	0.0	5	7.8	16.0	1.0
Discharged/transfer red/referre...	0.0	14	5.93	12.36	1.64
Neonate discharged to another ...	0.0	6	6.17	12.67	0.67
Discharged/transfer red/referre...	0.0	11	4.0	9.91	2.0

Discharge Disposition Risk Profile Sorted by Readmission Rate

Challenges & Limitations

✓ Challenges Solved

Patient-level data leakage

GroupShuffleSplit on patient_nbr → no patient in both train and test

Class imbalance (~11.2%)

SMOTE inside ImbPipeline; AUC & F1 over accuracy

ICD-9 cardinality (~900 codes)

Regex grouping into 17 clinical categories

Threshold calibration

Default 0.5 → tuned to ~0.15 ($\approx$ class prevalence)

⚠ Open Limitations

AUC capped ~0.62

Likely missing social-determinants-of-health features

RF + SMOTE interaction

Gini-based splits overfit synthetic minority samples

Data vintage (1999–2008)

May not reflect current care patterns or coding

Default threshold not deployable

Requires hospital-specific cost-ratio calibration

Conclusions & Future Work

What We Built

- Patient-level leak-safe pipeline
- 3 models · SMOTE-in-CV · threshold-calibrated
- Best AUC 0.625 (LR), F1 0.255 at threshold 0.135
- All 3 hypotheses rejected ($p < 0.0001$)

Key Insight

- Prior utilization $>$ any single lab value
- SNF transfers ($n \approx 14,000$) are the highest-impact target
- At threshold ~ 0.15 , flag $\sim 43\%$ of true readmissions
- 1 in 5–6 flagged patients are true readmissions

Future Work

- Add social-determinants-of-health features
- SHAP values for per-patient clinical explanation
- Fairness audit across age, race, payer subgroups
- Calibrate threshold to hospital-specific cost ratio